Funwork 2

(P1)

$$A_1 = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix} \cdot \begin{bmatrix} 1 & -1 \end{bmatrix}$$

$$\therefore \begin{bmatrix} 1 & -1 \\ 2 & -2 \\ -3 & 3 \end{bmatrix}$$

linear transform $A_1 : \mathbb{R}^2 \to \mathbb{R}^3$

Null space $N(A_1)$ Solve for $A_1 x = 0 \quad \therefore A_1 \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$

or $\begin{bmatrix} 1 & -1 \\ 2 & -2 \\ -3 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow x_1 - x_2 = 0 \quad \therefore x_1 = x_2$

$$\therefore N(A_1) = \text{span}\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}.$$

Range space $R(A_1)$

$$R(A_1) = \text{span}\left\{ \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix} \right\}$$, second col = -1 times first col

Rank $(A_1) = 1$ since two cols $c_1 = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$ & $c_2 = \begin{bmatrix} -1 \\ -2 \\ 3 \end{bmatrix}$ & $c_2 = -c_1$

so only one independent column

A_1 one-to-one? **No**

test $A_1 x = 0$ to have only trivial soln $x = 0$

$$\begin{bmatrix} 1 & -1 \\ 2 & -2 \\ -3 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 - x_2 \\ 2x_1 - 2x_2 \\ -3x_1 + 3x_2 \end{bmatrix} = 0. \quad \therefore x_1 - x_2 = 0 \Rightarrow x_1 = x_2$$

So infinite solutions $\therefore$ **not one-to-one**

another way $N(A_1) \neq \{0\}$ non zero vector

also (rank $(A_1) = 1$ and $< n = 2$)

A_1 onto? **No**

$R(A_1) \neq \mathbb{R}^3$ & (rank $(A_1) = 1 < m = 3$)

not onto

$$A_2 = \begin{bmatrix} 1 & 2 & -1 & 3 \\ -1 & 0 & -1 & 1 \\ -1 & -1 & 0 & -1 \end{bmatrix} \quad 3\times 4 \qquad A_2 : \mathbb{R}^4 \to \mathbb{R}^3$$

Row reduce : $R_2 \to R_2 + R_1$, $R_3 \to R_3 + R_1$, ~~$R_2 \to R_2/2$~~

$$A_{2\text{ref}} = \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & 2 & -2 & 4 \\ 0 & 1 & -1 & 2 \end{bmatrix} \xrightarrow{R_2 \to R_2/2} \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & 1 & -1 & 2 \\ 0 & 1 & -1 & 2 \end{bmatrix}$$

$$\downarrow R_3 \to R_3 - R_2$$

$$\begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \to R_1 - 2R_2} \begin{bmatrix} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\therefore A_{2RREF} = \begin{bmatrix} \boxed{1} & 0 & 1 & -1 \\ 0 & \boxed{1} & -1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Two pivots

$\therefore \text{rank}(A_2) = \underline{2}$.

To get $N(A_2)$ we set $A_{2\text{rref}} \cdot x = 0$

$$\therefore \quad x_1 + x_3 - x_4 = 0 \;, \quad x_1 = -x_3 + x_4$$

$$x_2 - x_3 + 2x_4 = 0 \;, \quad x_2 = x_3 - 2x_4$$

$$\therefore \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = x_3 \begin{bmatrix} -1 \\ 1 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 1 \\ -2 \\ 0 \\ 1 \end{bmatrix}$$

$$N(A_2) = \text{span}\left\{ \begin{bmatrix} -1 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -2 \\ 0 \\ 1 \end{bmatrix} \right\}$$

$x_1 = -s + t$
$x_2 = s - 2t$
$x_3 = s$
$x_4 = t$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -s+t \\ s-2t \\ s \\ t \end{bmatrix}$$

Range space $R(A_2)$ spanned by two pivot columns

$$\text{Range}(A_2) = \text{span}\left\{ \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} \right\}$$

Is A_2 one to one? No

Since $N(A_2) \neq \{0\}$ and rank $(A_2) = 2 < n = 4$

A_2 is not one to one

Is A_2 onto? No

$R(A_2)$ is 2D subspace of $\mathbb{R}^3$ $\therefore R(A_2) \neq \mathbb{R}^3$

and rank $(A_2) = 2 < m = 3$.

A_2 is not onto.

(P2)

$$A_c = \begin{bmatrix} -a_1 & -a_2 & \cdots & -a_{n-1} & -a_n \\ 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & & 0 & 0 \\ \vdots & & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{bmatrix}$$

State dynamic of controller canonical form

To prove,

(a) Characteristic polyn of A_c

$\det(\lambda I - A_c)$ is $\lambda^n + a_1\lambda^{n-1} + \cdots a_{n-1}\lambda + a_n$.

$$\lambda I - A_c = \begin{bmatrix} \lambda + a_1 & a_2 & a_3 & \cdots & a_{n-1} & a_n \\ -1 & \lambda & 0 & \cdots & 0 & 0 \\ 0 & -1 & \lambda & \ddots & 0 & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & -1 & \lambda & 0 \\ 0 & 0 & \cdots & 0 & -1 & \lambda \end{bmatrix}_{n \times n} \quad \text{(1)}$$

when $n = 1$,

$A_c = [-a_1]$ $\quad \lambda I_1 - A_c = \lambda + a_1$

$\therefore X_1(\lambda) = \det(\lambda I_1 - A_c) = \lambda + a_1$

$n = 2$ $\quad \lambda I_2 - A_c = \begin{bmatrix} \lambda + a_1 & a_2 \\ -1 & \lambda \end{bmatrix}$ $\quad \therefore X_2(\lambda) = \lambda^2 + a_1\lambda + a_2$

$n = 3$, $A_c = \begin{bmatrix} -a_1 & -a_2 & -a_3 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$, $\lambda I_3 - A_c$

$$= \begin{bmatrix} \lambda + a_1 & a_2 & a_3 \\ -1 & \lambda & 0 \\ 0 & -1 & \lambda \end{bmatrix}$$

$X_3(\lambda) = \det(\lambda I_3 - A_c)$

$$= (\lambda + a_1) \begin{vmatrix} \lambda & 0 \\ -1 & \lambda \end{vmatrix} - a_2 \begin{vmatrix} -1 & 0 \\ 0 & \lambda \end{vmatrix} + a_3 \begin{vmatrix} -1 & \lambda \\ 0 & -1 \end{vmatrix}$$

$$= (\lambda + a_1)\lambda^2 + a_2\lambda + a_3 = \lambda^3 + a_1\lambda^2 + a_2\lambda + a_3$$

$n = 4$ $\lambda I_4 - A_c = \begin{bmatrix} \lambda + a_1 & a_2 & a_3 & a_4 \\ -1 & \lambda & 0 & 0 \\ 0 & -1 & \lambda & 0 \\ 0 & 0 & -1 & \lambda \end{bmatrix}$

$$\therefore \det(\lambda I_4 - A_c) = \begin{vmatrix} \lambda + a_1 & a_2 & a_3 & a_4 \\ -1 & \lambda & 0 & 0 \\ 0 & -1 & \lambda & 0 \\ 0 & 0 & -1 & \lambda \end{vmatrix}$$

$$= \lambda + a_1 \underbrace{\begin{vmatrix} \lambda & 0 & 0 \\ -1 & \lambda & 0 \\ 0 & -1 & \lambda \end{vmatrix}}_{M_{11}} - a_2 \underbrace{\begin{vmatrix} -1 & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & -1 & \lambda \end{vmatrix}}_{M_{12}} + a_3 \underbrace{\begin{vmatrix} -1 & \lambda & 0 \\ 0 & -1 & 0 \\ 0 & 0 & \lambda \end{vmatrix}}_{M_{13}} + a_4 \underbrace{\begin{vmatrix} -1 & \lambda & 0 \\ 0 & -1 & \lambda \\ 0 & 0 & -1 \end{vmatrix}}_{M_{14}} \quad \text{—— (2)}$$

M_{11}

$$\begin{vmatrix} \lambda & 0 & 0 \\ -1 & \lambda & 0 \\ 0 & -1 & \lambda \end{vmatrix} = \lambda(\lambda^2) = \lambda^3$$

M_{12}

$$\begin{vmatrix} -1 & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & -1 & \lambda \end{vmatrix} = -1(\lambda^2) = -\lambda^2$$

M_{13}

$$\begin{vmatrix} -1 & \lambda & 0 \\ 0 & -1 & 0 \\ 0 & 0 & \lambda \end{vmatrix} = -1(-\lambda) - \lambda(0) + 0 = \lambda$$

M_{14}

$$\begin{vmatrix} -1 & \lambda & 0 \\ 0 & -1 & \lambda \\ 0 & 0 & -1 \end{vmatrix} = -1(1) - \lambda(0) + 0 = -1$$

(2) becomes

$$= (\lambda + a_1)\lambda^3 + a_2\lambda^2 + a_3\lambda + a_4$$

$$= \lambda^4 + a_1\lambda^3 + a_2\lambda^2 + a_3\lambda + a_4$$

So $\chi_1(\lambda) = \lambda + a_1$

$\chi_2(\lambda) = \lambda^2 + a_1\lambda + a_2$

$\chi_3(\lambda) = \lambda^3 + a_1\lambda^2 + a_2\lambda + a_3$

$\chi_4(\lambda) = \lambda^4 + a_1\lambda^3 + a_2\lambda^2 + a_3\lambda + a_4$

from base cases $n=1$ to $n=4$,

trend is $\chi_n(\lambda) = \lambda^n + a_1\lambda^{n-1} + a_2\lambda^{n-2} + \ldots + a_{n-1}\lambda + a_n$

To prove using induction hypothesis we assume it works for $n-1$ & then use to prove it works for n as inductive step

eqn ① for $\lambda I_n - A_c =$

$$\begin{bmatrix} \lambda + a_1 & a_2 & a_3 & \cdots & a_{n-1} & a_n \\ -1 & \lambda & \cdots & & 0 & 0 \\ 0 & -1 & \lambda & \cdots & 0 & 0 \\ 0 & & \cdots & -1 & \lambda & 0 \\ 0 & \cdots & \cdots & \cdots & -1 & \lambda \end{bmatrix}_{n\times n}$$

last term non zero entries two ie. a_n & λ

∴ Using Cofactor formular of nth col:

$$\det(A) = \sum_{i=1}^{n} a_{in}(-1)^{i+n}\det(M_{in})$$

$$\det(\lambda I_n - A_c) = a_n(-1)^{1+n}\det(M_{1n}) + \lambda(-1)^{n+n}\det(M_{nn})$$

$$= a_n(-1)^{1+n}\det(M_{1n}) + 1.\lambda\det(M_{nn})$$

$$M_{1n} = \begin{bmatrix} -1 & \lambda & \cdots & 0 \\ 0 & -1 & \lambda & \cdots \\ \vdots & & \ddots & \lambda \\ 0 & \cdots & & -1 \end{bmatrix}$$ upper triangular

$\therefore \det(M_{1n}) = (-1)^{n-1}$ | prod of $n-1$ diagonal entries

$\therefore a_n(-1)^{1+n}\det(M_{1n})$

$= a_n(-1)^{1+n}\cdot(-1)^{n-1}$

$= a_n(-1)^{1+n+n-1} = a_n.$

Second term

$\lambda \det(M_{nn})$

$$M_{nn} = \begin{vmatrix} \lambda + a_1 & \cdots & a_{n-1} \\ -1 & \lambda & 0 \\ & -1 \cdots & \\ & & \lambda \end{vmatrix} \quad \dim \; n-1$$

Since we assumed its valid for $n-1$ by induction hypothesis

$$\therefore \det(M_{nn}) = \lambda^{n-1} + a_1 \lambda^{n-2} + a_2 \lambda^{n-3} + \cdots + a_{n-2}\lambda + a_{n-1}$$

$$\therefore \lambda \det(M_{nn}) = \lambda^n + a_1 \lambda^{n-1} + a_2 \lambda^{n-2} + \cdots + a_{n-2}\lambda^2 + a_{n-1}\lambda$$

$$\therefore \det(\lambda I_n - A_c) = a_n + \lambda^n + a_1 \lambda^{n-1} + a_2 \lambda^{n-2} + \cdots + a_{n-2}\lambda^2 + a_{n-1}\lambda$$

$$\therefore \det(\lambda I_n - A_c) = \lambda^n + a_1 \lambda^{n-1} + a_2 \lambda^{n-2} + \cdots + a_{n-1}\lambda + a_n$$

Q.E.D. ∎

P2

(b) $A = \begin{bmatrix} 1 & 1 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}_{3\times 3}$

from part (a) we can get expression of (part n=3) for 3×3

$$\therefore \det(\lambda I_3 - A) = \begin{vmatrix} \lambda + a_1 & a_2 & a_3 \\ -1 & \lambda & 0 \\ 0 & -1 & \lambda \end{vmatrix} = \lambda^3 + a_1\lambda^2 + a_2\lambda + a_3 \quad \text{---(1)}$$

from companion matrix,

$$-a_1 = 1, \quad a_1 = -1, \quad -a_2 = 1 = 1, \quad a_2 = -1$$

$$-a_3 = -1, \quad a_3 = 1.$$

plugging in (1),

$$\det(\lambda I_3 - A) = \lambda^3 + (-1)\cdot\lambda^2 + (-1)\lambda + 1$$

$$= \lambda^3 - \lambda^2 - \lambda + 1 = \lambda^2(\lambda - 1) - (\lambda - 1)$$

$$= (\lambda^2 - 1)(\lambda - 1)$$

$$= (\lambda - 1)^2(\lambda + 1)$$

Solving $\chi_A(\lambda) = \det(\lambda I_3 - A) = 0$

$$(\lambda-1)^2(\lambda+1) = 0$$

$\lambda_1 = 1, 1$ (A.M = 2)

$\lambda_2 = -1$ (A.M = 1)

$\therefore$ eigen values are $\lambda = 1, 1, -1$ Answer

P2 (c)

To find,

$K = [k_1 \ k_2 \ k_3]$

s.t eigen val of $A - BK$ is $\{-1, -1, -1\}$

$$A = \begin{bmatrix} 1 & 1 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \qquad B = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$BK = \underset{3\times1}{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}} \underset{1\times3}{[k_1 \ k_2 \ k_3]} = \underset{3\times3}{\begin{bmatrix} k_1 & k_2 & k_3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}}$$

$$A - BK = \begin{bmatrix} 1-k_1 & 1-k_2 & -1-k_3 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

Using companion matrix,

$-a_1 = 1 - k_1$, $a_1 = k_1 - 1$

$-a_2 = 1 - k_1$, $a_2 = k_2 - 1$

$-a_3 = 1 - k_3$, $a_3 = 1 + k_3$

from (1) we have for 3×3 matrix

$$\chi_{A-BK}(\lambda) = \lambda^3 + a_1\lambda^2 + a_2\lambda + a_3$$

$$= \lambda^3 + \underbrace{(k_1-1)}_{a_1}\lambda^2 + \underbrace{(k_2-1)}_{a_2}\lambda + \underbrace{(1+k_3)}_{a_3} \quad \text{——(1)}$$

[redacted]

Let the three eigen values $\lambda_1, \lambda_2, \lambda_3$

characteristic polynomial $\chi(\lambda) = (\lambda-\lambda_1)(\lambda-\lambda_2)(\lambda-\lambda_3)$

Since eigenvalues are $-1, -1, -1$ $\therefore \chi(\lambda) = (\lambda+1)^3 = \lambda^3 + 3\lambda^2 + 3\lambda + 1$ ——(2)

∴ Comparing ① & ② coefficients

$k_1 - 1 = 3$, $k_2 - 1 = 3$, $k_3 + 1 = 1$

∴ $k_1 = 4$, $k_2 = 4$, $k_3 = 0$

∴ $k = [4, 4, 0]$ Ans

(P3) $A = \begin{bmatrix} -1 & 2 & -1 \\ 2 & -2 & -2 \\ -3 & 2 & 1 \end{bmatrix}$

(a) Characteristic eqn: $\chi(\lambda) = \det(\lambda I - A)$

$$\lambda I - A = \begin{bmatrix} \lambda+1 & -2 & 1 \\ -2 & \lambda+2 & 2 \\ 3 & -2 & \lambda-1 \end{bmatrix}$$

$$\det(\lambda I - A) = \begin{vmatrix} \lambda+1 & -2 & 1 \\ -2 & \lambda+2 & 2 \\ 3 & -2 & \lambda-1 \end{vmatrix}$$

$$= (\lambda+1)\{(\lambda+2)(\lambda-1)+4\} + 2\{(-2\lambda+2) - (3\lambda+6)\}$$

$$+ 4 - (3\lambda+6)$$

$$= (\lambda+1)(\lambda^2+2)(\lambda-1) + 4(\lambda+1) - 4\lambda + 4 - 12 + 4 - 3\lambda - 6$$

$$= \lambda^3 + 2\lambda^2 + 3\lambda + 2 - 4\lambda - 8 - 3\lambda - 2$$

$$= \lambda^3 + 2\lambda^2 - 4\lambda - 8$$

∴ Characteristic eqn is $\lambda^3 + 2\lambda^2 - 4\lambda - 8 = 0$ Ans

$= \lambda^2(\lambda+2) - 4(\lambda+2)$ $= (\lambda+2)(\lambda^2-4)$

$= (\lambda+2)^2(\lambda-2)$

$\chi(\lambda) = \det(\lambda I - A) = 0$ ∴ $(\lambda+2)^2(\lambda-2) = 0$.

$\lambda_1 = 2$, $\lambda_2 = -2$, ~~(A.M = 2)~~

$\lambda = 2$ (A.M = 1) & $\lambda = -2$ (A.M = 2) } eigenvalues with algebraic multiplicity. Ans

here $A \in \mathbb{R}^{3\times 3}$ is diagonalizable if $\exists$ invertible $T \in \mathbb{C}^{n\times n}$ s.t. $T^{*}AT = \Lambda$, Λ is diagonal $= \text{diag}(\lambda_1, \ldots \lambda_n)$

Since $\lambda = 2$ has A.M = 1, the G.M is 1 too at least

so we start with $\lambda = -2$ to find

$$N(A - (-2)I) = N(A + 2I)$$

$$A = \begin{bmatrix} -1 & 2 & -1 \\ 2 & -2 & -2 \\ -3 & 2 & 1 \end{bmatrix}$$

$$A + 2I = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 0 & -2 \\ -3 & 2 & 3 \end{bmatrix} \xrightarrow{R_2 \to \frac{1}{2}R_2} \begin{bmatrix} 1 & 2 & -1 \\ 1 & 0 & -1 \\ -3 & 2 & 3 \end{bmatrix} \xrightarrow{R_1 \to R_1 - R_2} \begin{bmatrix} 0 & 2 & 0 \\ 1 & 0 & -1 \\ -3 & 2 & 0 \end{bmatrix}$$

$$\xrightarrow{R_3 \to R_3 + 3R_2} \begin{bmatrix} 0 & 2 & 0 \\ 1 & 0 & -1 \\ 0 & 2 & 0 \end{bmatrix} \xrightarrow{R_3 \to R_3 - R_1} \begin{bmatrix} 0 & 2 & 0 \\ 1 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \rightleftarrows R_2} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

RREF form

$\therefore$ rank = 2 $\therefore$ two pivots $\therefore$ dim $N(A+2I) = 3 - \text{rank}$

$= 3 - 2 = 1$.

Here if $A \in \mathbb{R}^{3\times 3}$ is diagonalizable it should have 3 linearly independent eigenvectors. But $\lambda = -2$ gives eigenspace of dim = 1 so there is only one eigenvector associated with -2. with $\lambda = 2$, there is another eigenvector. So A only have two linearly independent eigenvector so A is not diagonalizable. Ans

P3

(b) $\sin(A)$ & A^{1000} via approach of remainder poly.

From part (a) we got characteristic eqn poly of A

$$\chi(\lambda) = \lambda^3 + 2\lambda^2 - 4\lambda - 8$$

Using C-H theorem, $f(\lambda) = q(\lambda) \cdot \chi(\lambda) + h(\lambda)$

quotient; remainder poly with degree $n-1$

here $h(\lambda)$ has degree $n-1 = 3-1 = 2$

$$\therefore h(\lambda) = a_0 + a_1\lambda + a_2\lambda^2$$

$\chi_A(A) = 0$ by C-H, so it reduced to,

$$\therefore f(A) = h(A) = a_0 I + a_1 A + a_2 A^2$$

$\lambda_1 = 2$ (a.m = 1)

$h(2) = f(2)$ or $a_0 + 2a_1 + 4a_2 = f(2)$ — ①

$\lambda_2 = -2$ (a.m = 2)

$h(-2) = f(-2)$ or $a_0 - 2a_1 + 4a_2 = f(-2)$ — ②

$h'(-2) = f'(-2)$ or $a_1 - 4a_2 = f'(-2)$ — ③

eqn ① − ②: $a_1 = \dfrac{f(2) - f(-2)}{4}$

a_1 in eqn ③: $-4a_2 = f'(-2) - \left(\dfrac{f(2) - f(-2)}{4}\right) = \dfrac{4f'(-2) - f(2) + f(-2)}{4}$

$$a_2 = \frac{f(2) - f(-2) - 4f'(-2)}{16}$$

$$\therefore a_0 = f(-2) + 2a_1 - 4a_2$$

$$= f(-2) + \frac{f(2) - f(-2)}{2} - \frac{f(2) - f(-2) - 4f'(-2)}{4}$$

$$= \frac{f(2) + 3f(-2)}{4} + f'(-2)$$

$f(\lambda) = \sin\lambda$, $f'(\lambda) = \cos\lambda$

$f(2) = \sin(2)$, $f(-2) = \sin(-2) = -\sin(2)$

$f'(-2) = \cos(-2) = +\cos(2)$

$$\therefore a_1 = \frac{\sin(2) - (-\sin(2))}{4} = \frac{1}{2}\sin(2)$$

$$a_2 = \frac{+\sin(2) + \sin(2) - 4\cos(2)}{16} = \frac{2\sin(2) - 4\cos(2)}{16}$$

$$= \frac{1}{8}\sin(2) - \frac{1}{4}\cos(2)$$

$$a_0 = \frac{\sin(2) + 3(-\sin(2) + 4\cdot\cos(2)}{4}$$

$$= -\frac{1}{2}\sin(2) + \cos(2)$$

Substituting a_0, a_1 & a_2 in $h(\lambda)$

$$\therefore h(\lambda) = \left(-\frac{1}{2}\sin(2) + \cos(2)\right) + \left(\frac{1}{2}\sin(2)\right)\lambda + \left(\frac{1}{8}\sin(2) - \frac{1}{4}\cos(2)\right)\lambda^2$$

$\therefore \sin(A) = h(A)$

$$\therefore \sin(A) = \left(\frac{1}{8}\sin(2) - \frac{1}{4}\cos(2)\right)A^2 + \frac{1}{2}\sin(2)\cdot A + \left(\cos(2) - \frac{1}{2}\sin(2)\right)I \quad \text{—— (4)}$$

$$A = \begin{bmatrix} -1 & 2 & -1 \\ 2 & -2 & -2 \\ -3 & 2 & 1 \end{bmatrix}, \quad A^2 = \begin{bmatrix} -1 & 2 & -1 \\ 2 & -2 & -2 \\ -3 & 2 & 1 \end{bmatrix}\begin{bmatrix} -1 & 2 & -1 \\ 2 & -2 & -2 \\ -3 & 2 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1+4+3 & -2-4-2 & 1-4-1 \\ -2-4+6 & 4+4+4 & -2+4-2 \\ 3+4-3 & -6-4+2 & 3-4+1 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 8 & -8 & -4 \\ 0 & 4 & 0 \\ 4 & -8 & 0 \end{bmatrix}$$

$$\therefore \sin A = \sin(2)\left[-\tfrac{1}{2}I + \tfrac{1}{2}A + \tfrac{1}{8}A^2\right] + \cos(2)\cdot\left[I - \tfrac{1}{4}A^2\right]$$ (12)

factor $\left(I - \frac{1}{4}A^2\right) = \begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix} - \frac{1}{4}\begin{bmatrix}8&-8&-4\\0&4&0\\4&-8&0\end{bmatrix}$

$$= \begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix} - \begin{bmatrix}2&-2&-1\\0&1&0\\1&-2&0\end{bmatrix}$$

$$= \begin{bmatrix}-1&2&1\\0&0&0\\-1&2&1\end{bmatrix}$$

factor $\left(-\frac{1}{2}I + \frac{1}{2}A + \frac{1}{8}A^2\right) = \begin{bmatrix}-1/2&0&0\\0&-1/2&0\\0&0&-1/2\end{bmatrix} + \begin{bmatrix}-1/2&1&-1/2\\1&-1&-1\\-3/2&1&1/2\end{bmatrix}$ b_3]

$$+ \begin{bmatrix}1&-1&-1/2\\0&1/2&0\\1/2&-1&0\end{bmatrix}$$

$$= \begin{bmatrix}0&0&-1\\1&-1&-1\\-1&0&0\end{bmatrix}$$

$$\therefore \sin(A) = \sin(2)\begin{bmatrix}0&0&-1\\1&-1&-1\\-1&0&0\end{bmatrix} + \cos(2)\begin{bmatrix}-1&2&1\\0&0&0\\-1&2&1\end{bmatrix}$$

$$= \begin{bmatrix}-\cos(2) & 2\cos(2) & -\sin(2)+\cos(2)\\ \sin(2) & -\sin(2) & -\sin(2)\\ -\sin(2)-\cos(2) & 2\cos(2) & \cos(2)\end{bmatrix}$$

Any

UGH!

Now A^{1000}

$f(\lambda) = \lambda^{1000}$, $f'(\lambda) = 1000\,\lambda^{999}$

$f(2) = 2^{1000}$, $f(-2) = (-2)^{1000} = 2^{1000}$, $f'(-2) = 1000\,(-2)^{999}$

$= -1000 \cdot 2^{999}$

$= -500 \cdot 2^{1000}$

$$\therefore \quad a_1 = \frac{f(2) - f(-2)}{4} = \frac{2^{1000} - 2^{1000}}{4} = 0$$

$$a_2 = \frac{2^{1000} - 2^{1000} - 4\left(-500 \cdot 2^{1000}\right)}{16} = \frac{1}{4} \cdot 500 \cdot 2^{1000}$$

$$= 125 \cdot 2^{1000}$$

$$a_0 = \frac{f(2) + 3f(-2) + f'(-2)}{4}$$

$$= \frac{2^{1000} + 3(2^{1000}) - 500 \cdot 2^{1000}}{4} = 2^{1000} - 500 \cdot 2^{1000}$$

$$= -499\, 2^{1000}$$

$$\therefore h(\lambda) = 125\, 2^{1000} \lambda^2 - 499\, 2^{1000}$$

$$= 2^{1000}\left(125\lambda^2 - 499\right)$$

$$\therefore A^{1000} = h(A) = 2^{1000}\left(125A^2 - 499I\right)$$

$$125A^2 = 125 \begin{bmatrix} 8 & -8 & -4 \\ 0 & 4 & 0 \\ 4 & -8 & 0 \end{bmatrix} = \begin{bmatrix} 1000 & -1000 & -500 \\ 0 & 500 & 0 \\ 500 & -1000 & 0 \end{bmatrix}$$

$$499I = \begin{bmatrix} 499 & 0 & 0 \\ 0 & 499 & 0 \\ 0 & 0 & 499 \end{bmatrix}$$

$$125A^2 - 499I = \begin{bmatrix} 501 & -1000 & -500 \\ 0 & 1 & 0 \\ 500 & -1000 & -499 \end{bmatrix}$$

$$\therefore A^{1000} = 2^{1000} \begin{bmatrix} 501 & -1000 & -500 \\ 0 & 1 & 0 \\ 500 & -1000 & -499 \end{bmatrix}$$

Ans

P3(c)

```
A = sym([-1 2 -1;
    2 -2 -2;
    -3 2 1]);

[P,J] = jordan(A)
```

P =

$$\begin{pmatrix} \frac{1}{2} & -1 & \frac{1}{2} \\ \frac{1}{2} & 0 & -\frac{1}{2} \\ -\frac{1}{2} & -1 & \frac{1}{2} \end{pmatrix}$$

J =

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & -2 & 1 \\ 0 & 0 & -2 \end{pmatrix}$$

```
P = sym([1/2 -1 1/2;
         1/2  0 -1/2;
        -1/2 -1 1/2]);

Pinv = inv(P);

simplify(Pinv*A*P)
```

ans =

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & -2 & 1 \\ 0 & 0 & -2 \end{pmatrix}$$

```
P = sym([1/2, -1,  1/2;
    1/2,  0, -1/2;
    -1/2, -1,  1/2]);

SinJ = [ sin(sym(2)), 0,            0;
    0,          -sin(sym(2)), cos(sym(2));
    0,           0,           -sin(sym(2))];

SinA = simplify(P * SinJ * inv(P));

disp(SinA)
```

$$\begin{pmatrix} -\cos(2) & 2\cos(2) & \cos(2) - \sin(2) \\ \sin(2) & -\sin(2) & -\sin(2) \\ -\cos(2) - \sin(2) & 2\cos(2) & \cos(2) \end{pmatrix}$$

```
P = sym([1/2, -1,  1/2;
```

```
    1/2,  0, -1/2;
    -1/2, -1,  1/2]);

Pinv = inv(P);

Jk = [2^k, 0, 0;
    0, (-2)^k, k*(-2)^(k-1);
    0, 0, (-2)^k];

Ak = simplify(P * Jk * Pinv);

disp(Ak)
```

$$\begin{pmatrix} \sigma_2+\sigma_1+\frac{2^k}{2} & -(-2)^k k & \sigma_1-\sigma_2-\frac{2^k}{2} \\ \frac{2^k}{2}-\sigma_1 & (-2)^k & \sigma_1-\frac{2^k}{2} \\ \sigma_2+\sigma_1-\frac{2^k}{2} & -(-2)^k k & \sigma_1-\sigma_2+\frac{2^k}{2} \end{pmatrix}$$

where

$$\sigma_1=\frac{(-2)^k}{2}$$

$$\sigma_2=\frac{(-2)^k k}{2}$$

$$A = \begin{bmatrix} -1 & 2 & -1 \\ 2 & -2 & -2 \\ -3 & 2 & 1 \end{bmatrix}$$

P3

(c) From matLab, 1×1 block $\lambda = 2$, $J_1 = [2]$, 2×2 for $\lambda = -2$, $J_2 = \begin{bmatrix} -2 & 1 \\ 0 & -2 \end{bmatrix}$

To satisfy

$$J = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -2 & 1 \\ 0 & 0 & -2 \end{bmatrix}$$

$$A = PJP^{-1}$$

or $AP = PJ$

$$P = \begin{bmatrix} 1/2 & -1 & 1/2 \\ 1/2 & 0 & -1/2 \\ -1/2 & -1 & 1/2 \end{bmatrix}$$

from Matlab (allowed?)

$p_1 \quad p_2 \quad p_3$

let $P = [p_1 \; p_2 \; p_3]$

$PJ = [2p_1 \; 2p_2 \; p_2 - 2p_3]$

$AP = PJ$

verify

$\therefore Ap_1 = 2p_1$

$Ap_2 = -2p_2$

$Ap_3 = p_2 - p_3$

$(A - 2I)p_1 = 0, \quad (A + 2I)p_2 = 0, \quad (A + 2I)p_3 = p_2$

$\lambda = 2$

$(A - 2I)p_1 = 0$ or $A - 2I = \begin{bmatrix} -3 & 2 & -1 \\ 2 & -4 & -2 \\ -3 & 2 & -1 \end{bmatrix}$

$R_3 \to R_1 - R_3$ $\begin{bmatrix} -3 & 2 & -1 \\ 2 & -4 & -2 \\ 0 & 0 & 0 \end{bmatrix}$

$R_1 \leftrightarrow R_2$, $R_1 \to \frac{1}{2}R_1$

$\begin{bmatrix} 2 & -4 & -2 \\ -3 & 2 & -1 \\ 0 & 0 & 0 \end{bmatrix}$

$R_1 \to \frac{1}{2}R_1$ $\begin{bmatrix} 1 & -2 & -1 \\ -3 & 2 & -1 \\ 0 & 0 & 0 \end{bmatrix}$

$R_2 \to R_2 + 3R_1$ $\begin{bmatrix} 1 & -2 & -1 \\ 0 & -4 & -4 \\ 0 & 0 & 0 \end{bmatrix}$

$R_2 \to -\frac{1}{4}R_2$ $\begin{bmatrix} 1 & -2 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

$R_1 \to R_1 + 2R_2$ $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ RREF

$x + z = 0, \quad y + z = 0$

if $p_1 = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

$\therefore x = -z$

$y = -z$

$p_1 = z \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} +1/2 \\ +1/2 \\ -1/2 \end{bmatrix}$

$\Rightarrow Ap_1 = 2p_1$

$\lambda = -2$

$(A+2I)p_2 = 0$, $(A+2I) = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 0 & -2 \\ -3 & 2 & 3 \end{bmatrix}$

$\downarrow$ $R_2 \to R_2 - 2R_1$, $R_3 \to R_3 + 3R_1$

$\begin{bmatrix} 1 & 2 & -1 \\ 0 & -4 & 0 \\ 0 & 8 & 0 \end{bmatrix}$ $\xleftarrow{R_2 \to -\frac{1}{4}R_2}$ $\begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 0 \\ 0 & 8 & 0 \end{bmatrix}$ $\xleftarrow{R_1 \to R_1 - 2R_2,\ R_3 \to R_3 - 8R_2}$ $\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

RREF form $\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$p_2 = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

$x - z = 0$, $y = 0$

$x = z$, $y = 0$

$p_2 = z \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

$= 0 \begin{bmatrix} -1 \\ 0 \\ -1 \end{bmatrix}$

$\therefore Ap_2 = -2p_2$

$\lambda = -2$, $J_2 = \begin{bmatrix} -2 & 1 \\ 0 & -2 \end{bmatrix}$

$(A+2I)p_3 = p_2 = \begin{bmatrix} -1 \\ 0 \\ -1 \end{bmatrix}$

$\therefore \begin{bmatrix} 1 & 2 & -1 \\ 2 & 0 & -2 \\ -3 & 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ -1 \end{bmatrix}$

$x + 2y - z = -1$

$2x - 2z = 0$

$-3x + 2y + 3z = -1$

$x = z$,

$\therefore 2y = -1$

$y = -1/2$

$\therefore p_3 = \begin{bmatrix} 1/2 \\ -1/2 \\ 1/2 \end{bmatrix}$

$\Rightarrow Ap_3 = p_2 - 2p_3$

$P = [p_1 \quad p_2 \quad p_3]$

$P = \begin{bmatrix} 1/2 & -1 & 1/2 \\ 1/2 & 0 & -1/2 \\ -1/2 & -1 & 1/2 \end{bmatrix}$ hence matches matlab output.

$\det P = \frac{1}{2}\left(0 - \frac{1}{2}\right) + 1\left(\frac{1}{4} - \frac{1}{4}\right) + \frac{1}{2}\left[-\frac{1}{2}\right]$

$= -1/4 - 1/4 = -2/4 = -1/2 \neq 0$ hence invertible

P^{-1} by gauss Jordan. let $Q = 2P$

$$Q = \begin{bmatrix} 1 & -2 & 1 \\ 1 & 0 & -1 \\ -1 & -2 & 1 \end{bmatrix}$$

$$P = \frac{1}{2}Q \qquad P^{-1} = 2Q^{-1}$$

$$\left[\begin{array}{ccc|ccc} 1 & -2 & 1 & 1 & 0 & 0 \\ 1 & 0 & -1 & 0 & 1 & 0 \\ -1 & -2 & 1 & 0 & 0 & 1 \end{array}\right] \xrightarrow[R_3 \to R_3 + R_1]{R_2 \to R_2 - R_1} \left[\begin{array}{ccc|ccc} 1 & -2 & 1 & 1 & 0 & 0 \\ 0 & 2 & -2 & -1 & 1 & 0 \\ 0 & -4 & 2 & 1 & 0 & 1 \end{array}\right]$$

$\downarrow R_2 \to \frac{1}{2} R_2$

$$\left[\begin{array}{ccc|ccc} 1 & -2 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & -\frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 & \frac{1}{2} & -1 & -\frac{1}{2} \end{array}\right] \xleftarrow{R_3 \to -\frac{1}{2} R_3} \left[\begin{array}{ccc|ccc} 1 & -2 & 1 & 1 & 0 & 0 \\ 0 & 2 & -2 & -1 & 1 & 0 \\ 0 & -4 & 2 & 1 & 0 & 1 \end{array}\right]$$

$\downarrow R_2 \to R_2 + R_3$, $R_1 \to R_1 - R_3$

$$\left[\begin{array}{ccc|ccc} 1 & -2 & 0 & \frac{1}{2} & 1 & \frac{1}{2} \\ 0 & 1 & 0 & 0 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} & -1 & -\frac{1}{2} \end{array}\right] \xrightarrow{R_1 \to R_1 + 2R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{2} & 0 & -\frac{1}{2} \\ 0 & 1 & 0 & 0 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} & -1 & -\frac{1}{2} \end{array}\right]$$

$$\therefore Q^{-1} = \begin{bmatrix} \frac{1}{2} & 0 & -\frac{1}{2} \\ 0 & -\frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -1 & -\frac{1}{2} \end{bmatrix} \qquad P^{-1} = 2Q^{-1}$$

$$\therefore P^{-1} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & -1 \\ 1 & -2 & 1 \end{bmatrix}$$

$$\therefore A = PJP^{-1} = \begin{bmatrix} \frac{1}{2} & -1 & \frac{1}{2} \\ \frac{1}{2} & 0 & -\frac{1}{2} \\ -\frac{1}{2} & -1 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & -2 & 1 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & -1 \\ 1 & -2 & 1 \end{bmatrix}$$

$\therefore$ To compute for $\sin(A)$

$$\sin(A) = P \sin(J) P^{-1}$$

For 2×2 Jordan block $J_2 = \begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix}$ the matrix function is $f(J_2) = \begin{bmatrix} f(\lambda) & f'(\lambda) \\ 0 & f(\lambda) \end{bmatrix}$

for $f(x) = \sin x$

$f(-2) = \sin(-2) = -\sin(2)$, $f'(-2) = \cos(-2) = \cos(2)$

$f(2) = \sin(2)$

$f(a) = \sin a$

$$\therefore \sin(J_2) = \begin{bmatrix} -\sin(2) & \cos 2 \\ 0 & -\sin(2) \end{bmatrix}$$

full Jordan block

$$\therefore \sin(J) = \begin{bmatrix} \sin(2) & 0 & 0 \\ 0 & -\sin(2) & \cos(2) \\ 0 & 0 & -\sin(2) \end{bmatrix}$$

$$\sin(A) = \begin{bmatrix} \frac{1}{2} & -1 & \frac{1}{2} \\ \frac{1}{2} & 0 & -\frac{1}{2} \\ -\frac{1}{2} & -1 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} \sin(2) & 0 & 0 \\ 0 & -\sin(2) & \cos(2) \\ 0 & 0 & -\sin(2) \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & -1 \\ 1 & -2 & -1 \end{bmatrix}$$

$\updownarrow$

or

$$\sin(A) = \begin{bmatrix} -\cos(2) & 2\cos(2) & \cos(2) - \sin(2) \\ \sin(2) & -\sin(2) & -\sin(2) \\ -\cos(2) - \sin(2) & 2\cos(2) & \cos(2) \end{bmatrix}$$

Ans

→ Matlab.

Next for A^k $\quad A^2 = P J^2 P^{-1}$, $A^3 = P J^3 P^{-1}$ .. $A^k = P J^k P^{-1}$

$J_1 = [2]^k = [2^k]$, $J_2 = \begin{bmatrix} -2 & 1 \\ 0 & -2 \end{bmatrix}$ =

Using $f(J_2) = \begin{bmatrix} f(a) & f'(a) \\ 0 & f(a) \end{bmatrix}$

$f(a) = a^k$, $f'(a) = k a^{k-1}$

$a = -2$ for J_2 $\quad \therefore J_2^k = \begin{bmatrix} (-2)^k & k(-2)^{k-1} \\ 0 & (-2)^k \end{bmatrix}$

$$\therefore \text{full } J^k = \begin{bmatrix} 2^k & 0 & 0 \\ 0 & (-2)^k & k(-2)^{k-1} \\ 0 & 0 & (-2)^k \end{bmatrix}$$

$\therefore A^k = P J^k P^{-1}$ substituting

$$A^k = \begin{bmatrix} 1/2 & -1 & 1/2 \\ 1/2 & 0 & -1/2 \\ -1/2 & -1 & 1/2 \end{bmatrix} \begin{bmatrix} 2^k & 0 & 0 \\ 0 & (-2)^k & k(-2)^{k-1} \\ 0 & 0 & (-2)^k \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & -1 \\ 1 & -2 & 1 \end{bmatrix}$$

Ans

or from Matlab

$$A^k = \begin{bmatrix} \frac{2^k + (k+1)(-2)^k}{2} & -(-2)^k k & \frac{(1-k)(-2)^k - 2^k}{2} \\ \frac{2^k - (-2)^k}{2} & (-2)^k & \frac{(-2)^k - 2^k}{2} \\ \frac{(k+1)(-2)^k - 2^k}{2} & -k(-2)^k & \frac{2^k + (1-k)(-2)^k}{2} \end{bmatrix}$$

Ans

— × —

P4 (a) $A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$ ⊛ $A^T = A$ so it's diagonalizable.

To find e^{At} find first char poly & Eigenvals

$$\chi_A(\lambda) = \det(\lambda I - A) = \det \begin{bmatrix} \lambda & -1 \\ -1 & -1+\lambda \end{bmatrix}$$

$$= \lambda(-1+\lambda) - 1 \quad = -\lambda + \lambda^2 - 1$$

$$\lambda = \frac{+1 \pm \sqrt{1+4}}{2} = \frac{1 \pm \sqrt{5}}{2}$$

$$\therefore \lambda_1 = \frac{1+\sqrt{5}}{2} \quad \& \quad \lambda_2 = \frac{1-\sqrt{5}}{2} \qquad \left.\begin{array}{l} \lambda_1 + \lambda_2 = 1 \\ \lambda_1 \lambda_2 = -1 \end{array}\right\}$$

two linearly independent 2 eigen vectors that are distinct ∴ A is diagonalizable

eigenvalue $(\lambda_i I - A) v_i = 0 \qquad \begin{bmatrix} \lambda_i & -1 \\ -1 & \lambda - 1 \end{bmatrix} \begin{bmatrix} x_i \\ y_i \end{bmatrix} = 0$

$$-\lambda_1 x_i + y_i = 0 \qquad y_i = \lambda_1 x_i \qquad \text{say } x_1 = 1$$

$$v_i = \begin{bmatrix} 1 \\ \lambda_i \end{bmatrix}$$

$$\therefore v_1 = \begin{bmatrix} 1 \\ \lambda_1 \end{bmatrix} = \begin{bmatrix} 1 \\ \frac{1+\sqrt{5}}{2} \end{bmatrix}$$

$$v_2 = \begin{bmatrix} 1 \\ \lambda_2 \end{bmatrix} = \begin{bmatrix} 1 \\ \frac{1-\sqrt{5}}{2} \end{bmatrix}$$

diagonalization $A = P \Lambda P^{-1}$

$$P = \begin{bmatrix} 1 & 1 \\ \lambda_1 & \lambda_2 \end{bmatrix}, \quad \Lambda = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}, \quad \det(P) = \lambda_2 - \lambda_1$$

$$\therefore P^{-1} = \frac{1}{\lambda_2 - \lambda_1} \begin{bmatrix} \lambda_2 & -1 \\ -\lambda_1 & 1 \end{bmatrix}$$

$$= \frac{1}{\frac{1-\sqrt{5}}{2} - \frac{-1-\sqrt{5}}{2}} \begin{bmatrix} \lambda_2 & -1 \\ -\lambda_1 & 1 \end{bmatrix} = -\frac{1}{\sqrt{5}} \begin{bmatrix} \lambda_2 & -1 \\ -\lambda_1 & 1 \end{bmatrix}$$

$$= \frac{1}{\sqrt{5}} \begin{bmatrix} -\lambda_2 & 1 \\ \lambda_1 & -1 \end{bmatrix}$$

$e^{At} = P e^{\Lambda t} P^{-1}$

$$e^{At} = P \begin{bmatrix} e^{\lambda_1 t} & 0 \\ 0 & e^{\lambda_2 t} \end{bmatrix} P^{-1}$$

$$= \frac{1}{\sqrt{5}} \begin{bmatrix} 1 & 1 \\ \lambda_1 & \lambda_2 \end{bmatrix} \begin{bmatrix} e^{\lambda_1 t} & 0 \\ 0 & e^{\lambda_2 t} \end{bmatrix} \begin{bmatrix} -\lambda_2 & 1 \\ \lambda_1 & -1 \end{bmatrix}$$

$$= \frac{1}{\sqrt{5}} \begin{bmatrix} e^{\lambda_1 t} & e^{\lambda_2 t} \\ \lambda_1 e^{\lambda_1 t} & \lambda_2 e^{\lambda_2 t} \end{bmatrix} \begin{bmatrix} -\lambda_2 & 1 \\ \lambda_1 & -1 \end{bmatrix}$$

$$= \frac{1}{\sqrt{5}} \begin{bmatrix} -\lambda_2 e^{\lambda_1 t} + \lambda_1 e^{\lambda_2 t} & e^{\lambda_1 t} - e^{\lambda_2 t} \\ -\lambda_1 \lambda_2 e^{\lambda_1 t} + \lambda_1 \lambda_2 e^{\lambda_2 t} & \lambda_1 e^{\lambda_1 t} - \lambda_2 e^{\lambda_2 t} \end{bmatrix}$$

or since $\lambda_1 \lambda_2 = -1$

$$= \frac{1}{\sqrt{5}} \begin{bmatrix} -\lambda_2 e^{\lambda_1 t} + \lambda_1 e^{\lambda_2 t} & e^{\lambda_1 t} - e^{\lambda_2 t} \\ e^{\lambda_1 t} - e^{\lambda_2 t} & \lambda_1 e^{\lambda_1 t} - \lambda_2 e^{\lambda_2 t} \end{bmatrix} \quad \text{for } t \geq 0$$

Ans

P4 (b) Laplace Method :-

$e^{At} = \mathcal{L}^{-1}\{(sI - A)^{-1}\}$:

$$sI - A = \begin{bmatrix} s & -1 \\ -1 & s-1 \end{bmatrix} \quad | \quad (sI-A)^{-1} \det(sI - A)$$

$$= s(s-1) - 1$$

$$= s^2 - s - 1$$

$$(sI - A)^{-1} = \frac{1}{s^2 - s - 1} \begin{bmatrix} s-1 & 1 \\ 1 & s \end{bmatrix}$$

$$\therefore (SI-A)^{-1} = \begin{bmatrix} \frac{s-1}{(s-\lambda_1)(s-\lambda_2)} & \frac{1}{(s-\lambda_1)(s-\lambda_2)} \\ \frac{1}{(s-\lambda_1)(s-\lambda_2)} & \frac{s}{(s-\lambda_1)(s-\lambda_2)} \end{bmatrix}$$

now $\mathcal{L}^{-1}\{(SI-A)^{-1}\}$, we do entry by entry of each scalar

$$\therefore e^{At} = \mathcal{L}^{-1}\{(SI-A)^{-1}\} = \begin{bmatrix} \mathcal{L}^{-1}\left\{\frac{s-1}{(s-\lambda_1)(s-\lambda_2)}\right\} & \mathcal{L}^{-1}\left\{\frac{1}{(s-\lambda_1)(s-\lambda_2)}\right\} \\ \mathcal{L}^{-1}\left\{\frac{1}{(s-\lambda_1)(s-\lambda_2)}\right\} & \mathcal{L}^{-1}\left\{\frac{s}{(s-\lambda_1)(s-\lambda_2)}\right\} \end{bmatrix}$$

$\lambda_1\lambda_2 = -1$

$\lambda_1 + \lambda_2 = 1$

$\lambda_2 - \lambda_1 = \sqrt{5}$

off diagonal entries each case.

$$\frac{1}{(s-\lambda_1)(s-\lambda_2)} = \frac{C}{(s-\lambda_1)} + \frac{D}{(s-\lambda_2)}$$

$$1 = C(s-\lambda_2) + D(s-\lambda_1)$$

choosing $s = \lambda_1$ (λ_1) $\Rightarrow$ $C = \frac{1}{\lambda_1-\lambda_2} = -\frac{1}{\sqrt{5}}$

when $s = \lambda_2$ (λ_2) $\Rightarrow$ $D = \frac{1}{\lambda_2-\lambda_1} = \frac{1}{\sqrt{5}}$

$$\therefore \frac{1}{(s-\lambda_1)(s-\lambda_2)} = \frac{-1}{\sqrt{5}}\frac{1}{s-\lambda_1} + \frac{1}{\sqrt{5}}\frac{1}{s-\lambda_2}$$

$$= \frac{1}{\sqrt{5}}\left[\frac{1}{s-\lambda_2} - \frac{1}{s-\lambda_1}\right]$$

Since, $\mathcal{L}^{-1}\left\{\frac{1}{s-a}\right\} = e^{at}$

$$\therefore \mathcal{L}^{-1}\left\{\frac{1}{(s-\lambda_1)(s-\lambda_2)}\right\} = \frac{1}{\sqrt{5}}\left[e^{\lambda_2 t} - e^{\lambda_1 t}\right]$$

now upper left (1,1). $\mathcal{L}^{-1}\left\{\frac{s-1}{(s-\lambda_1)(s-\lambda_2)}\right\}$

$$\frac{s-1}{(s-\lambda_1)(s-\lambda_2)} = \frac{C}{s-\lambda_1} + \frac{D}{s-\lambda_2}$$

P4(c)

```
syms t

A = sym([0 1;
    1 1]);

eAt = simplify(expm(A*t));

disp(eAt)
```

$$\begin{pmatrix} \frac{\sigma_2 \left(5\,e^{\sqrt{5}t} - \sqrt{5}\,e^{\sqrt{5}t} + \sqrt{5} + 5\right)}{10} & \sigma_1 \\ \sigma_1 & \frac{\sigma_2 \left(5\,e^{\sqrt{5}t} + \sqrt{5}\,e^{\sqrt{5}t} - \sqrt{5} + 5\right)}{10} \end{pmatrix}$$

where

$$\sigma_1 = \frac{\sqrt{5}\,\sigma_2 \left(e^{\sqrt{5}t} - 1\right)}{5}$$

$$\sigma_2 = e^{-\frac{t\,(\sqrt{5}-1)}{2}}$$

$\therefore (s-1) = C(s-a_2) + D(s-a_1)$

choose $s = a_1$, $\therefore a_1 - 1 = C(a_1 - a_2)$

$-a_2 = C(a_1 - a_2)$ | $a_1 + a_2 = 1$, $a_1 - 1 = -a_2$

$= C(-\sqrt{5})$

$\therefore C = \frac{a_2}{\sqrt{5}}$

$s = a_2 \Rightarrow (a_2 - 1) = D(a_2 - a_1) \quad \therefore D = -\frac{a_1}{\sqrt{5}}$

$-a_1 = D(\sqrt{5})$

$\therefore \frac{s-1}{(s-a_1)(s-a_2)} = \frac{a_2}{\sqrt{5}} \frac{1}{s-a_1} - \frac{a_1}{\sqrt{5}} \frac{1}{s-a_2}$

$\mathcal{L}^{-1}\left\{\frac{s-1}{(s-a_1)(s-a_2)}\right\} = \mathcal{L}^{-1}\left\{\frac{a_2}{\sqrt{5}} \frac{1}{s-a_1}\right\} - \mathcal{L}^{-1}\left\{\frac{a_1}{\sqrt{5}} \frac{1}{s-a_2}\right\}$

$= \frac{a_2}{\sqrt{5}} e^{a_1 t} - \frac{a_1}{\sqrt{5}} e^{a_2 t}$

$= \frac{a_2 e^{a_1 t} - a_1 e^{a_2 t}}{\sqrt{5}}$

lower right (2,2).

$$\frac{s}{(s-a_1)(s-a_2)} = \frac{C}{(s-a_1)} + \frac{D}{(s-a_2)}$$

or $s = C(s-a_2) + D(s-a_1)$

$s = a_1 \Rightarrow a_1 = C(s-a_2)$

$C = -\frac{1}{\sqrt{5}}$

$s = a_2 \Rightarrow D = a_2/\sqrt{5}$

$$\therefore \frac{s}{(s-a_1)(s-a_2)} = -\frac{a_1}{\sqrt{5}} \frac{1}{(s-a_1)} + \frac{a_2}{\sqrt{5}} \frac{1}{(s-a_2)}$$

$$\therefore \mathcal{L}^{-1}\left(\frac{s}{(s-a_1)(s-a_2)}\right) = -\frac{a_1}{\sqrt{5}} e^{a_1 t} + \frac{a_2}{\sqrt{5}} e^{a_2 t} = \frac{-a_1 e^{a_1 t} + a_2 e^{a_2 t}}{\sqrt{5}}$$

$$\therefore e^{At} = \frac{1}{\sqrt{5}} \begin{bmatrix} \lambda_2 e^{\lambda_1 t} - \lambda_1 e^{\lambda_2 t} & e^{\lambda_2 t} - e^{\lambda_1 t} \\ e^{\lambda_2 t} - e^{\lambda_1 t} & -\lambda_1 e^{\lambda_1 t} + \lambda_2 e^{\lambda_2 t} \end{bmatrix}$$

Ans

& $\lambda_1 = \frac{1-\sqrt{5}}{2}$ | $\lambda_2 = \frac{1+\sqrt{5}}{2}$

——×——

P5

P5 (a) Given $A = \begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix}$

Using C-H theorem remainder polynomial.

Characteristic polynomial $\chi_A(\lambda) = \det(\lambda I - A) = 0$

$$\lambda I - A = \begin{bmatrix} \lambda & 1 \\ -1 & \lambda+1 \end{bmatrix} \quad \therefore \chi_A(\lambda) = \lambda(\lambda+1) - (-1) = 0$$

$$= \lambda^2 + \lambda + 1 = 0$$

$$\therefore \lambda^2 + \lambda + 1 = 0 \Rightarrow \lambda_1 = \frac{-1+\sqrt{3}\,i}{2} \text{ & } \lambda_2 = \frac{-1-\sqrt{3}\,i}{2} \qquad i^2 = -1$$

Using C-H,

because the characteristic polynomial is of degree 2, the remainder polynomial can be at least (≤2) degree 1., so $e^{At} = C_0(t) I + C_1(t) A$ with $C_0(t)$ & $C_1(t)$ scalar fn of t. ----①

and, first degree remainder poly is

$$h(\lambda) = C_0(t) + C_1(t)\lambda$$

if λ_j eigen value of A, $\chi_A(\lambda_j) = 0$ by defn. | $h(\lambda_j) = f(\lambda_j)$

$\therefore$ at $\lambda = \lambda_j \Rightarrow e^{\lambda_j t} = C_0(t) + C_1(t)\lambda_j$

$$\lambda_1 = \overset{Re}{\sigma} + \overset{Im}{i\omega} \qquad \lambda_2 = \overset{Re}{\sigma} - \overset{Im}{i\omega}$$

$\sigma = -1/2$, $\omega = \sqrt{3}/2$

$\therefore \lambda_1 = \sigma + i\omega$ $\quad i^2 = -1$

$\lambda_2 = \sigma - i\omega$

P5 (a) Cont...

$$c_0(t) + c_1(t)(\sigma + i\omega) = e^{(\sigma+i\omega)t} \quad -$$

Since $e^{i\theta} = \cos\theta + i\sin\theta$

$$e^{(\sigma+i\omega)t} = e^{\sigma t}[\cos(\omega t) + i\sin(\omega t)]$$

Im part

$$c_0(t) + \sigma c_1(t) + i\omega c_1(t) = e^{\sigma t}\cos(\omega t) + i e^{\sigma t}\sin(\omega t)$$

$$\omega c_1(t) = e^{\sigma t}\sin(\omega t)$$

$$c_1(t) = \frac{e^{\sigma t}\sin(\omega t)}{\omega} = \frac{e^{-t/2}\sin\left(\frac{\sqrt{3}}{2}t\right)}{\sqrt{3}/2}$$

Re part

$$c_0(t) + \sigma c_1(t) = e^{\sigma t}\cos(\omega t)$$

$$c_0(t) = e^{\sigma t}\cos(\omega t) - \sigma c_1(t)$$

$$c_0(t) = e^{-\frac{1}{2}t}\cos\left(\frac{\sqrt{3}}{2}t\right) + \frac{1}{2}\left[\frac{2}{\sqrt{3}}e^{-t/2}\sin\left(\frac{\sqrt{3}}{2}t\right)\right]$$

$$c_0(t) = e^{-t/2}\left[\cos\left(\frac{\sqrt{3}}{2}t\right) + \frac{1}{\sqrt{3}}\sin\left(\frac{\sqrt{3}}{2}t\right)\right]$$

Add in eqn (1)

$$e^{At} = c_0(t)\,I + c_1(t)\,A = \begin{bmatrix} c_0(t) & 0 \\ 0 & c_0(t) \end{bmatrix} + \begin{bmatrix} 0 & -c_0(t) \\ c_1(t) & -c_1(t) \end{bmatrix}$$

$$\therefore e^{At} = \begin{bmatrix} c_0(t) & -c_0(t) \\ c_1(t) & c_0(t) - c_1(t) \end{bmatrix}$$

now substituting $c_1(t)$ & $c_0(t)$ in the matrix above

wait!

$$c_0(t) - c_1(t) = e^{-t/2}\left[\cos\left(\frac{\sqrt{3}}{2}t\right) + \frac{1}{\sqrt{3}}\sin\left(\frac{\sqrt{3}}{2}t\right) - \frac{2}{\sqrt{3}}\sin\left(\frac{\sqrt{3}}{2}t\right)\right.$$

$$= e^{-t/2}\left[\cos\left(\frac{\sqrt{3}}{2}t\right) - \frac{1}{\sqrt{3}}\sin\left(\frac{\sqrt{3}}{2}t\right)\right]$$

$$\therefore e^{At} = e^{-t/2}\begin{bmatrix} \cos\left(\frac{\sqrt{3}}{2}t\right) + \frac{1}{\sqrt{3}}\sin\left(\frac{\sqrt{3}}{2}t\right) & -\frac{2}{\sqrt{3}}\sin\left(\frac{\sqrt{3}}{2}t\right) \\ \frac{2}{\sqrt{3}}\sin\left(\frac{\sqrt{3}}{2}t\right) & \cos\left(\frac{\sqrt{3}}{2}t\right) - \frac{1}{\sqrt{3}}\sin\left(\frac{\sqrt{3}}{2}t\right) \end{bmatrix}$$

Ans

2nd Method

Using Laplace $\therefore e^{At} = \mathcal{L}^{-1}\{(sI - A)^{-1}\}$

$$sI - A = \begin{bmatrix} s & 1 \\ -1 & s+1 \end{bmatrix}$$

$$\det(sI - A) = (s^2 + s + 1)$$

$$= s^2 + s + 1$$

$$= s^2 + 2.\frac{1}{2}s + \frac{1}{4} + \frac{3}{4}$$

$$= \left(s + \frac{1}{2}\right)^2 + \frac{3}{4}$$

$$= \left(s + \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2$$

$$\therefore (sI - A)^{-1} = \frac{1}{s^2 + s + 1}\begin{bmatrix} s+1 & -1 \\ 1 & s \end{bmatrix}$$

$\therefore$ ~~$\mathcal{L}^{-1}(sI - A)$~~

Top left (1,1)

$$\frac{s+1}{\left(s+\frac{1}{2}\right)^2 + \left(\sqrt{3}/2\right)^2} = \frac{s + 1/2 + 1/2}{(\cdot\cdot)^2 + (\cdot\cdot)^2} = \frac{s + 1/2}{\left(s+\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} + \frac{\frac{1}{2}}{\left(s+\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}$$

Using,

$$\mathcal{L}^{-1}\left\{\frac{s}{s^2 + a^2}\right\} = \cos(at) \quad \& \quad \mathcal{L}\{e^{-ct}f(t)\} = F(s+c)$$

$$\therefore \mathcal{L}\{e^{-ct}\cos(at)\} = \frac{s+c}{(s+c)^2 + a^2}$$

$$\mathcal{L}(\sin(at)) = \frac{a}{s^2 + a^2}$$

$$\mathcal{L}\{e^{-ct}\sin at\} = \frac{a}{(s+c)^2 + a^2}$$

$$\mathcal{L}^{-1}\left\{\frac{s + (1/2)}{\left(s + (1/2)\right)^2 + \left(\sqrt{3}/2\right)^2}\right\} = e^{-t/2}\cos\left(\frac{\sqrt{3}}{2}t\right)$$

$$\mathcal{L}^{-1}\left\{\frac{1/2}{\left(s+\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{\sqrt{3}}\,\frac{\sqrt{3}/2}{\left(s+\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}\right\}$$

$$= \frac{1}{\sqrt{3}}e^{-t/2}\sin\left(\frac{\sqrt{3}}{2}t\right)$$

$\therefore$ entry (1,1)

$$\mathcal{L}^{-1}(\quad) = e^{-t/2}\cos\left(\frac{\sqrt{3}}{2}t\right) + \frac{1}{\sqrt{3}}e^{-t/2}\sin\left(\frac{\sqrt{3}}{2}t\right)$$

off diogonal one's .(1,2)

$$\mathcal{L}^{-1}\left\{\frac{-1}{(s+\frac{1}{2})^2+(\frac{\sqrt{3}}{2})^2}\right\} = -\frac{2}{\sqrt{3}}\,\frac{\sqrt{3}/2}{(s+1/2)^2+(\frac{\sqrt{3}}{2})^2} = -\frac{2}{\sqrt{3}}e^{-t/2}\sin\left(\frac{\sqrt{3}}{2}t\right)$$

& (2,1) $\mathcal{L}^{-1}\left\{\frac{1}{(s+\frac{1}{2})^2+(\sqrt{3}/2)^2}\right\} = \frac{2}{\sqrt{3}}te^{-t/2}\sin\left(\frac{\sqrt{3}}{2}t\right)$

for (2,2)

$$\mathcal{L}^{-1}\left\{\frac{s}{(s+\frac{1}{2})^2+(\frac{\sqrt{3}}{2})^2}\right\} = \mathcal{L}^{-1}\left\{\frac{s+1/2}{(s+1/2)^2+(\frac{\sqrt{3}}{2})^2}\right\} - \mathcal{L}^{-1}\left\{\frac{\frac{1}{\sqrt{3}}\cdot\sqrt{3}/2}{(s+\frac{1}{2})^2+(\frac{\sqrt{3}}{2})^2}\right\}$$

$$= e^{-t/2}\cos\left(\frac{\sqrt{3}}{2}t\right) - \frac{1}{\sqrt{3}}e^{-t/2}\sin\left(\frac{\sqrt{3}}{2}t\right)$$

$$e^{At} := \mathcal{L}^{-1}(sI-A)^{-1} = \frac{1}{s^2+s+1}\begin{bmatrix} s+1 & -1 \\ 1 & s \end{bmatrix}$$

$$e^{At} = e^{-t/2}\begin{bmatrix} \cos\left(\frac{\sqrt{3}}{2}t\right)+\frac{1}{\sqrt{3}}\sin\left(\sqrt{3}/2\,t\right) & -\frac{2}{\sqrt{3}}\sin\left(\frac{\sqrt{3}}{2}t\right) \\ \frac{2}{\sqrt{3}}\sin\left(\frac{\sqrt{3}}{2}t\right) & \cos\left(\frac{\sqrt{3}}{2}t\right)-\frac{1}{\sqrt{3}}\sin\left(\frac{\sqrt{3}}{2}t\right) \end{bmatrix}$$

Ans

P5

(b) By C-H theorem is my choice.

$$\gamma(a) = a^k = q(a)\,\chi_A(a) + h(a)$$

($h(a)$: remainder.)

second degree polynomial so remainder at most 1 degree

$$\therefore h(a) = c_0(k) + c_1(k)\,\lambda \quad \text{—①}$$

$$\therefore A^k \equiv h(A) = c_0(k)\,I + c_1(k)\,A. \quad \text{—②}$$

λ from $X_A(\lambda) = 0$ was.

$$\lambda_1 = \frac{-1+\sqrt{3}i}{2} \quad \& \quad \lambda_2 = \frac{-1-\sqrt{3}i}{2}$$

$$r = \sqrt{(-1/2)^2 + (\sqrt{3}/2)^2} = \sqrt{1/4 + 3/4} = 1 \qquad r = 1.$$

$$\cos\theta = -1/2 \qquad \sin\theta = \sqrt{3}/2 \qquad \therefore \theta = \frac{2\pi}{3}$$

$$\therefore \lambda_1 = \left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right) = e^{i2\pi/3}$$

$$\text{so, } \lambda_2 = e^{-i2\pi/3}$$

$$\therefore h(\lambda_i) = \lambda_i^k$$

$$\therefore \lambda_1^k = e^{i\frac{2\pi}{3}k} = \cos\frac{2\pi k}{3} + i\sin\left(\frac{2\pi k}{3}\right)$$

also, from eqn ①,

$$C_0(k) + C_1(k)\lambda_1 = C_0(k) + C_1(k)\left(-\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)$$

$$= \left[C_0(k) - \frac{1}{2}C_1(k)\right] + i\frac{\sqrt{3}}{2}C_1(k)$$

Re Im

$$\left[C_0(k) - \frac{1}{2}C_1(k)\right] + i\left[\frac{\sqrt{3}}{2}C_1(k)\right] = \cos\frac{2\pi k}{3} + i\sin\left(\frac{2\pi k}{3}\right)$$

$$\frac{\sqrt{3}}{2}C_1(k) = \sin\frac{2\pi k}{3}, \quad C_1(k) = \frac{2}{\sqrt{3}}\sin\left(\frac{2\pi k}{3}\right)$$

$$C_0(k) - \frac{1}{2}C_1(k) = \cos\left(\frac{2\pi k}{3}\right), \quad C_0(k) = \cos\left(\frac{2\pi k}{3}\right) + \frac{1}{2}C_1(k)$$

$$\therefore C_0(k) = \cos\left(\frac{2\pi k}{3}\right) + \frac{1}{\sqrt{3}}\sin\left(\frac{2\pi k}{3}\right)$$

from ②

$$\therefore A^k = C_0(k)I + C_1(k)A$$

$$= \begin{bmatrix} C_0(k) & 0 \\ 0 & C_0(k) \end{bmatrix} + \begin{bmatrix} 0 & -C_1(k) \\ C_1(k) & -C_1(k) \end{bmatrix}$$

$$= \begin{bmatrix} C_0(k) & -C_1(k) \\ C_1(k) & C_0(k) - C_1(k) \end{bmatrix} \quad \text{—— ③}$$

by subst^{n}

$$\therefore A^k = \begin{bmatrix} \cos\left(\frac{2\pi k}{3}\right) + \frac{1}{\sqrt{3}}\sin\left(\frac{2\pi k}{3}\right) & -\frac{2}{\sqrt{3}}\sin\left(\frac{2\pi k}{3}\right) \\ \frac{2}{\sqrt{3}}\sin\left(\frac{2\pi k}{3}\right) & \cos\left(\frac{2\pi k}{3}\right) - \frac{1}{\sqrt{3}}\sin\left(\frac{2\pi k}{3}\right) \end{bmatrix}$$

Ans

To check for $k = 0, 1, \ldots$? do I need to show?

$k = 0$, $A^k = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (as $\cos(0) = 1$, $\sin(0) = 0$)

$\therefore A^k = I$

$k = 1$ $\quad A^k = \begin{bmatrix} -\frac{1}{2} + \frac{1}{\sqrt{3}}\cdot\frac{\sqrt{3}}{2} & -\frac{2}{\sqrt{3}}\cdot\frac{\sqrt{3}}{2} \\ \frac{2}{\sqrt{3}}\cdot\frac{\sqrt{3}}{2} & -\frac{1}{2} - \frac{\sqrt{3}}{2}\cdot\frac{1}{\sqrt{3}} \end{bmatrix}$

$= \begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix} = A$ $\quad \therefore A^k = A$

$k = 2$ $\quad A^k = \begin{bmatrix} \left(-\frac{1}{2}\right) + \frac{1}{\sqrt{3}}\left(-\frac{\sqrt{3}}{2}\right) & -\frac{2}{\sqrt{3}}\cdot\left(-\frac{\sqrt{3}}{2}\right) \\ \frac{2}{\sqrt{3}}\cdot\left(-\frac{\sqrt{3}}{2}\right) & \left(-\frac{1}{2}\right) - \frac{1}{\sqrt{3}}\cdot\left(\frac{\sqrt{3}}{2}\right) \end{bmatrix}$

$A^k = \begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix} = A^2$

$$\therefore A^k = \begin{cases} I & k = 0 \\ A & k = 1 \\ A^2 & k = 2 \\ \cdots & \end{cases}$$

Am?

P5(a) optional

```
syms t

A = sym([0 -1;
    1 -1]);

eAt = simplify(expm(A*t));

disp(eAt)
```

$$\begin{pmatrix} \sigma_2 + \frac{\sigma_3}{2} - \frac{\sigma_1}{6} + \frac{\sqrt{3}\,\sigma_3\,\mathrm{i}}{6} & \frac{\sigma_1}{3} - \frac{\sqrt{3}\,\sigma_3\,\mathrm{i}}{3} \\ -\frac{\sigma_1}{3} + \frac{\sqrt{3}\,\sigma_3\,\mathrm{i}}{3} & \sigma_2 + \frac{\sigma_3}{2} + \frac{\sigma_1}{6} - \frac{\sqrt{3}\,\sigma_3\,\mathrm{i}}{6} \end{pmatrix}$$

where

$$\sigma_1 = \sqrt{3}\,\mathrm{e}^{\frac{t\,(-1+\sqrt{3}\,\mathrm{i})}{2}}\,\mathrm{i}$$

$$\sigma_2 = \frac{\mathrm{e}^{\frac{t\,(-1+\sqrt{3}\,\mathrm{i})}{2}}}{2}$$

$$\sigma_3 = \mathrm{e}^{-\frac{t\,(1+\sqrt{3}\,\mathrm{i})}{2}}$$